

NETWORKING 04

👉 What is an IP address?

Think like this:

- ➡ Your house has an address
- ➡ Your phone/laptop also has an **address on the internet**

That address = **IP Address**

📌 So:

IP = identity of a device on a network

IPv4 FORMAT (IMPORTANT)

You saw:

x.x.x.x

Example:

192.168.1.5

👉 Rules:

- 4 parts (called **octets**)
- Each part is from **0 to 255**

✓ Valid:

10.0.0.1
172.16.5.4
192.168.1.100

✗ Invalid:

300.1.1.1 (not allowed)



WHAT IS IPv6? (EASY UNDERSTANDING)

👉 IPv4 was like:

- Limited addresses (finished almost)

👉 So they created **IPv6 = next generation IP**

📌 Simple:

IPv6 = **new, bigger address system for internet devices**



FORMAT OF IPv6

Instead of numbers like IPv4, IPv6 looks like this:

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

😱 Looks scary? Break it:

👉 It has:

- 8 blocks
- Each block = 4 hexadecimal digits (0–9, a–f)



PRIVATE IP (VERY VERY IMPORTANT FOR EXAMS + FORENSICS)

These are **special IP ranges used inside homes, colleges, companies**

📌 Private IP ranges:

```
10.x.x.x  
172.16.x.x → 172.31.x.x  
192.168.x.x
```

😕 Why private?

Think like this:

- Your phone → gets private IP (inside WiFi)
- Your router → has public IP (internet facing)

👉 So your device is **hidden behind router**

🔥 This is called:

NAT (Network Address Translation)

If you see an IP in logs:

Case 1:

192.168.1.10

→ This is **private**

→ Means attack happened **inside network**

Case 2:

8.8.8.8

→ This is **public**

→ Means traffic came from **internet**

👉 **Is it private or public?**

Check:

IP starts with	Type
10	Private
172.16–31	Private
192.168	Private
Anything else	Public

👉 Example Answer (how you write in exam)

Your IP: 192.168.1.5

Interface: wlan0

Type: Private

Why:

Because it belongs to 192.168.x.x range, which is reserved for private networks.



REAL WORLD CYBER USE (IMPORTANT FOR YOU)

In **Digital Forensics / Security Jobs**, you will:

- Track attacker IPs in logs
- Identify if attack is:
 - Internal (employee)
 - External (hacker)
- Trace source using IP intelligence



1. NAT (Network Address Translation)



Simple idea:

👉 Many devices → ONE public IP

Example:

- Your phone → 192.168.1.5
- Your laptop → 192.168.1.6
- Router → **1 public IP (like 49.x.x.x)**

➡ Internet sees **ONLY** router IP



Forensic thinking:

👉 If logs show:

49.x.x.x

➡ Could be **100 devices** behind it

📌 So:

NAT = **hiding crowd behind one face**



2. VPN (Virtual Private Network)



Simple idea:

👉 You connect to another server → your IP becomes THEIR IP

Example:

- Your real IP → 49.36.x.x
- VPN server → 185.107.x.x

➡ Websites see:

185.107.x.x

🔒 Extra:

- Data is **encrypted**
- ISP cannot see activity clearly

🕶 3. PROXY



Simple idea:

👉 Middleman server

You → Proxy → Website

Example:

- You send request
- Proxy forwards it
- Website sees proxy IP



Difference from VPN:

Feature	VPN	Proxy
Encryption	Yes	Mostly No
Security	High	Low
Usage	Full system	Browser/app only



Forensic thinking:

- Easy proxies → easier to trace
- Logs may reveal real IP



HOW HACKERS HIDE IP

1 VPN

- Most common
- Changes IP

2 Proxy chains

- Multiple proxies

Attacker → Proxy1 → Proxy2 → Target

3 TOR (Dark web)

- 👉 Multiple layers of encryption
- 👉 Very hard to trace

4 Public WiFi attack

- Attacker uses café WiFi
 - IP belongs to café, not attacker

5 Compromised systems (Botnets)

- Hack someone else's PC
 - Attack from that system